

BlockBloom Project

Assignment 1

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1. List out 5 different use cases of blockchain technology. Mention at least one use case relevant to IITK Campus Community.
(Please be careful. Understand what exactly a blockchain achieves. Do not make the same mistakes you did in the recruitment task.)
2. List out 5 different blockchain networks in use. Write down what their speciality/specific use case is. What consensus mechanism do they use?
3. Write a program in any programming language that simulates the cryptographic problem solved in PoW. (Find the smallest nonce (a whole number) such that SHA-256 hash of the new string is less than the threshold.)
Input by user: String
Output: Nonce, Time taken to find the nonce
You can assume a threshold like 0x000FF.....F (that is the first 3 hexadecimal digits should be 0 for the nonce to be accepted). Also assume that the nonce has to be appended at the end of the given string.
4. Find out what is UTXO. Explain it in few words.
5. Why is a blockchain said to be immutable?
6. When a fraudulent block is added to a blockchain with PoW consensus mechanism by a criminal who does not have the ability to perform 51% attack, how is the fork resolved?
7. What is 'Nothing-at-stake' problem in PoS? How is this avoided?
8. Explain why 51% attack is less probable with PoS than with PoW.
9. What are digital signatures in the context of blockchains?
10. What is the Oracle Problem? How is it solved?
11. (Bonus/Optional) What are Zero-knowledge proofs? How are they used in the context of blockchains?

Deadline: 11:59 pm, 21 December 2024